

Minimal-Shot AV: External Transfer Diagnostic

Actor completion and selector removal do not erase the AlpaSim collision surface

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CoRL 2027 audit deck

CoRL-facing evidence

- Waymo AlpaSim matched transfer diagnostics
- WOD-E2E candidate-selection validation
- Per-axis logs: collision, offroad, wrong lane, progress, tracking

Excluded

Internal 2D simulator pass rates, COMPASS scores, and GIF rollouts are development artifacts only. They are not used as CoRL results in this deck or paper.

Collision persists after the two easiest explanations are supplied:

- **Actor visibility:** world-frame actor completion leaves raw collision unchanged.
- **Learned ranking:** learned-selector-free direct-grid replay improves only two collision clips.

The result is a localization result, not a positive safety claim.

AlpaSim Transfer Matrix

Policy	Collision	Offroad	Wrong lane	Progress
Token-DAGger	0.600	0.900	0.700	0.034
Clamped lateral	0.700	0.500	0.200	0.384
Axis-constrained clamped	0.700	0.200	0.200	0.358
Hard veto	0.800	0.200	0.700	0.837
Source decay	0.600	0.900	0.400	0.175

Each intervention repairs a different axis and leaves or worsens another. A single aggregate score would hide the failure mechanism.

Actor Completion Is Not Enough

Policy	Collision	Offroad	Wrong lane	Progress	Dist.-GT
Axis-constrained clamped	0.600	0.207	0.241	0.170	1.54
World-frame actor completion	0.600	0.310	0.172	0.184	1.42

Actor completion changes lane, offroad, progress, and tracking behavior, but does not reduce raw collision on the matched 30-scene rerun.

Selector Removal Is Not Enough

Policy	Collision	Offroad	Wrong lane	Progress
Clamped collision subset	1.000	0.118	0.235	0.167
Direct grid, cost-ranked	0.889	0.125	0.312	0.246
Direct grid, max-clearance	0.889	0.059	0.176	0.174

On the 18 baseline collision clips, bypassing the learned selector improves only two paired clips. The residual boundary is not just learned-ranking error.

WOD-E2E Grounding Placement

Ablation	RFS	Oracle	Regret
Scalar geometry	7.695	9.068	1.373
InternVLA only	7.672	9.225	1.554
Cosmos + InternVLA linear	7.728	9.208	1.480
Cosmos nonlinear head	7.845	9.264	1.419
Cosmos + world prior + direct selector	7.848	9.254	1.407

Frozen world priors help when they act at the candidate selector, not when appended as generic latent features.

The CoRL claim is narrow and auditable:

- Bounded candidate interfaces make transfer failures inspectable.
- Actor visibility and learned-selector removal are not sufficient.
- External diagnostics must report collision, offroad, lane, progress, and tracking separately.